

1. Complete Python Code (End-to-End)

Python

Import Libraries

```
import pandas as pd
```

```
import numpy as np
```

```
from sklearn.model_selection import train_test_split
```

```
from sklearn.linear_model import LogisticRegression
```

```
from sklearn.metrics import accuracy_score, precision_score, recall_score, confusion_matrix,  
classification_report
```

Load Dataset

```
df = pd.read_csv('/mnt/data/heart.csv')
```

Display basic info

```
print("Dataset Shape:", df.shape)
```

```
print(df.head())
```

Separate Features and Target

```
X = df.drop('target', axis=1) # Features
```

```
y = df['target']             # Target variable
```

Train-Test Split (80% train, 20% test)

```
X_train, X_test, y_train, y_test = train_test_split(  
    X, y, test_size=0.2, random_state=42
```

```
)
```

```
print("\nTraining Data Shape:", X_train.shape)
```

```
print("Testing Data Shape:", X_test.shape)
```

```
# Train Model (Logistic Regression)
```

```
model = LogisticRegression(max_iter=1000)
```

```
model.fit(X_train, y_train)
```

```
# Predictions
```

```
y_pred = model.predict(X_test)
```

```
# Evaluation Metrics
```

```
accuracy = accuracy_score(y_test, y_pred)
```

```
precision = precision_score(y_test, y_pred)
```

```
recall = recall_score(y_test, y_pred)
```

```
cm = confusion_matrix(y_test, y_pred)
```

```
# Results
```

```
print("\n--- Evaluation Metrics ---")
```

```
print("Accuracy:", accuracy)
```

```
print("Precision:", precision)
```

```
print("Recall:", recall)
```

```
print("\nConfusion Matrix:")
```

```
print(cm)
```

```
print("\nClassification Report:")
```

```
print(classification_report(y_test, y_pred))
```

2. Evaluation Report (Interpretation)

Accuracy

Measures overall correctness

Example: 0.85 → 85% predictions are correct

Precision

Out of predicted heart disease cases, how many are actually correct

High precision = fewer false positives

Recall

Out of actual heart disease cases, how many we correctly identified

High recall = fewer false negatives (very important in healthcare)

Confusion Matrix

Structure:

[[TN FP]

[FN TP]]

TN: Correctly predicted no disease

FP: Wrongly predicted disease

FN: Missed disease (dangerous)

TP: Correctly predicted disease

In medical problems, Recall is more important than Accuracy

(because missing a disease is risky)

3. Final Outcome

Dataset successfully split into training and testing sets

Logistic Regression model trained

Model evaluated using:

Accuracy

Precision

Recall

Confusion Matrix

Clear understanding of model performance

Intern understands model evaluation fundamentals

4. Interview Questions (With Answers)

What is Overfitting?

When a model learns training data too well (including noise) and performs poorly on new data.

Accuracy vs Precision?

Accuracy → Overall correctness

Precision → Correctness of positive predictions

Use precision when false positives matter

What is Recall?

Recall measures how many actual positives were correctly predicted.

Important in healthcare, fraud detection

What is Confusion Matrix?

A table that shows prediction results:

True Positive

True Negative

False Positive

False Negative

Why split data?

To evaluate model performance on unseen data and avoid overfitting.

Training set → learn patterns

Testing set → evaluate performances